How tumour classification works

Internal process, cut-offs and formulas, in plain language, each one tied to the measured result. Research and educational use only, not a medical diagnosis. The short results sheet for a project guide is guide_progress.pdf in the same folder.

1. The short answer

The program was asked: **is there a tumour**, **how big is it** and **where is it**. Every number comes from phantoms the program had never seen, in recording sessions where the session itself gives no hint. Two situations were tested: a **single scan** (a first-ever visit) and a **follow-up scan compared with earlier healthy scans of the same breast** (monitoring).

Question	Program	Recording details only	Simple guessing
Tumour? single scan (score /100)	71 / 100	47 / 100	50 / 100
Tumour? compared with earlier healthy scan	**97 / 100**	49 / 100	50 / 100
Alarm, single scan	66% found / 65% cleared	-	-
Alarm, compared with earlier scan	88% found / 95% cleared	-	-
Size class, single scan	64%	57%	33%
Size class, compared with earlier scan	56%	47%	33%
Where? gen3 image spot (no training)	**1.3 cm**	-	3.2 cm (the centre)

"/100": pick one tumour scan and one healthy scan at random; how often does the program give the tumour scan the higher score? 50 = coin flip, 100 = perfect. "Alarm": with the yes/no cut-off chosen on training phantoms only, the share of tumours found and of healthy scans correctly cleared. "Recording details only": a model that sees just session, antenna distance and generation - how much could be guessed without looking at the scan.

The official first-scan score is 71 / 100. That number is the nested test in section 3: the program was allowed to choose features, model and training pool on training phantoms only, then scored once. A later side check used only the signal features and only the fair training scans, and that narrower check scored about 69. It did not replace this result. Section 9 is that side check.

2. How the test was kept fair

• **No session shortcut.** Tumour and healthy phantoms were mostly scanned in different sessions, and the machine drifts. Only sessions with both kinds were used for testing, each trimmed to equal tumour and healthy counts (268 single-scan test scans from 27 phantoms). Recording details alone score about 50.

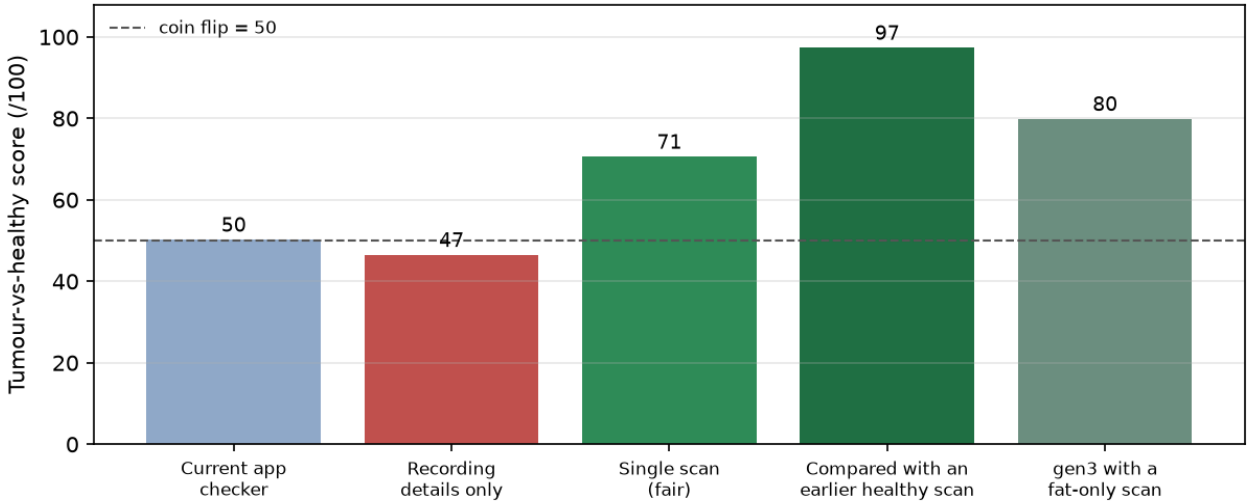
• **Never the same phantom in training and test.**

• **No picking the best of many tries.** Feature set, model type, training data and the alarm cut-off were chosen on the training phantoms only, then tested once ("nested" testing), repeated 3 times with different splits.

• **No help from the answer.** Tumour positions are never given to the program. Settings for the image (antenna layout, background removal) were chosen on one half of the phantoms and checked on the other half.

• **Earlier healthy scans are picked blindly.** Every scan, tumour or healthy, is compared with the same number of *other* healthy scans of the same phantom from the same session: as many as the scan that has the fewest, and at most 3. A tumour is never given a cleaner baseline than the healthy scans, so the count cannot give away the answer. Scans with only one earlier healthy scan are included.

3. Is there a tumour? Single scan



Single scan (fair test scans)	Result
Score /100 (all / gen1 / gen3)	71 ± 2 / 67 / 72
Alarm with the default cut-off (0.5)	74% found / 57% cleared
Alarm with the cut-off chosen on training phantoms	66% found / 65% cleared
Current app checker without training	50 / 100
Choice the program made most often	signal + image, random forest, trained on fair scans plus all other scans

From a single scan the tumour signal is real but weak. Choosing the cut-off properly balances the alarms but cannot add information that is not in the scan.

4. Comparing with an earlier healthy scan (monitoring)

The biggest problem is the background: skin, fat and inner tissue echo much more strongly than a tumour. If an earlier healthy scan of the same breast exists, subtracting it removes almost all of that background, leaving only what changed. Gen3 has several healthy scans of each phantom in the same session as its tumour scans, taken alternately. Gen1 is included too, down to phantoms that have only one healthy scan in that sitting. Both kinds of scan are handled the same way.

Is the change really the tumour? Bigger tumours should change the scan more; handling noise would not care about tumour size. Median amount of change in gen3 (relative to the scan's total signal):

Scan	Median change
Healthy (repeat scan)	2.0e-05
Tumour 1 cm	2.1e-05
Tumour 1.5 cm	4.0e-05
Tumour 2 cm	9.9e-05
Tumour 2.5 cm	1.6e-04
Tumour 3 cm	2.6e-04

The change grows steadily with tumour size, so it is caused by the tumour. It also shows the limit of this system: a 1 cm tumour changes the scan no more than repeating a healthy scan does.

Compared with earlier healthy scan (246 test scans, 24 phantoms)	Program	Recording details only	Guessing
Tumour? (score /100)	97 ± 0	49 / 100	50
Tumour? older / newest / 2 cm or smaller	86 / 99 / 97	-	-
Alarm (cut-off chosen on training phantoms)	88% found / 95% cleared	-	-
Size class (balanced accuracy, three sizes)	56%	47%	33%
Size class, newest phantoms only	90%	-	-
Diameter, average error	0.4 cm	0.7 cm	0.8 cm
Diameter, newest phantoms only	0.2 cm	-	-

Catch: this answers "did something new appear since the last healthy scan?", not "is there a tumour in a first-ever scan?". It needs a previous healthy scan of the same person in the same position. The difference image did not locate the tumour better, because small shifts of the inner tissue between scans also show up in it.

5. How big is it? (single scan)

Tumour scans of the fair set	Program	Recording details only	Guessing
Size class: small (≤2 cm) / medium (≤4 cm) / large	64%	57%	33%
Diameter, average error	0.7 cm	0.7 cm	0.9 cm

Most of the size skill comes from knowing the generation (gen1 used 2, 4 and 6 cm tumours, gen3 used 1-3 cm). The scan itself adds only a few points.

6. Where is it? The image step

The app's original image missed the tumour by more than simply guessing the centre. Three changes fixed it for gen3: remove the part of the signal that looks the same from every antenna angle (skin, fat shell, chamber); add the echoes up with their wave phase (coherent delay-and-sum); and fit the antenna layout per generation. Then several further background-removal ideas were compared (gen3, average miss; settings chosen on one half of the phantoms):

Background removal (gen3)	Average miss	Within 2 cm
Empty chamber + same-from-every-angle part removed, 1-8 GHz	1.6 cm	77%
Remove the strongest shared pattern	4.1 cm	0%
Remove the 2 strongest shared patterns	4.2 cm	25%
Remove the 3 strongest shared patterns	3.9 cm	13%
First row + cut echoes before 0.5 ns (skin)	1.6 cm	77%
First row + cut echoes before 1 ns	1.7 cm	76%
First row + cut echoes before 1.5 ns	1.7 cm	79%
First row, low band only (1-4 GHz)	1.3 cm	81%
First row, middle band only (2-6 GHz)	3.0 cm	52%
First row, high band only (4-8 GHz)	4.8 cm	6%

Using only the low frequencies (1-4 GHz) worked best and was picked by both halves; on the held-out half it misses by 1.3 cm against 3.2 cm for the centre. Removing the strongest shared patterns removed the tumour as well, and cutting the early skin echo changed nothing.



In gen1 and gen2 nothing beat guessing the centre: their dense inner tissue sits off-centre and outshines the tumour. Gen3 has only four tumour positions, so part of its skill is picking the right quarter.

Position (all fair single-scan tumours)	Average miss	Within 2 cm
Guess the centre	3.2 cm	1%
Image spot, no training (official)	1.8 cm	65%
Trained program	2.0 cm	53%
Recording details only (warning sign)	2.1 cm	50%

Why the trained program is not used for position: within a session the tumour sat in the same spot, so a model that sees only the session places it about as well. The image spot never sees labels or sessions, so it is the honest result.

7. Best case for a single scan: a fat-only scan (gen3)

Every gen3 phantom was also scanned with only its fat layer. A real patient never has such a scan, so this is a ceiling for single-scan background removal, not a usable method.

gen3, 200 scans, 20 phantoms	With fat-only scan
Tumour? (score /100)	80 ± 3
Alarm	73% found / 70% cleared
Size class (balanced accuracy)	81%
Diameter average error	0.3 cm (recording details only: 0.6 cm)

8. What this means

• **Single first-time scan:** tumours are detected better than chance but with many false alarms; not good enough for decisions. Subtracting an average of other breasts that share the same fat shell made this worse, so a first visit still uses the empty chamber only.

• **Follow-up against an earlier healthy scan:** this is the result to use. Small tumours (2 cm or less) score about 97/100, the newest phantoms about 99/100, and the older phantoms about 86/100.

• **Location:** works in gen3 with the new image steps (best with 1-4 GHz); not in gen1/gen2.

• **Size:** on a follow-up the newest phantoms are sized well. The older phantoms are not, and a 1 cm tumour still changes the raw scan about as much as repeating a healthy scan.

• **Generation 2** cannot be added. Its tumour scans and healthy scans were taken in different sittings, and the sitting number alone gives the answer away.

9. Mathematical analysis of a first visit (side check, not a new official score)

The official first-scan score stays the nested result in section 3. This section asks a different question: if the same forest is given only the signal samples, does adding the formulas below raise the score? The signal-only run of this check is about 69, because it is not allowed the image features or the extra training scans that the nested test used.

A first visit has no earlier healthy scan, so the skin and fat shell cannot be subtracted. After the empty-chamber scan is removed, the part that looks the same from every antenna is subtracted. What remains is summarised by three formulas, and those formulas are handed to the same forest that already reads the time samples.

Scattering spectrum. The breast-window signals form a matrix B, one column per antenna. Its singular values s give L_i = s_i^2 / (sum of s_j^2). L_1 is the share of energy in the strongest pattern, and the effective rank is 1 / (sum of L_i^2). A compact inclusion raises L_1. Tissue that fills the breast spreads the energy.

Frequency slope. p(f) is the average leftover magnitude. Over 2–6 GHz the fit is log10 p(f) = a + b log10(f / 1 GHz). A small sphere in the Born approximation has b near 2, because the scattered field grows with frequency squared. A larger inclusion pulls b down. b does not depend on how hard the antennas were driven, so generation 1 and generation 3 can share it.

Compactness. Excess kurtosis of the energy-weighted delay is high for a short echo and low for a smear. The angular resultant length R is high when one antenna direction carries the leftover.

Where, without training. On the brightest leftover antenna, d = R_antenna - c t_peak / 2, and the antenna angle supplies the direction. This is the radar round-trip, not a fitted image.

The comparison below was fixed before looking at the result: the same model, the same phantom-grouped folds (5 folds, 3 seeds, 268 fair first-visit scans), with and without the physics features. No other model was tried.

First visit	Signal only	Signal + physics	Physics formulas only
Tumour score /100	69 ± 2	68 ± 2	51 ± 2
Older phantoms (gen1)	56 ± 1	56 ± 2	45 ± 1
Newest phantoms (gen3)	74 ± 3	73 ± 2	54 ± 3
2 cm or smaller	63 ± 2	62 ± 2	47 ± 1
Found / cleared at the 0.5 cut	73 ± 2 / 53 ± 4	73 ± 1 / 53 ± 3	50 ± 2 / 51 ± 10

The formulas do not improve the first visit. Signal samples alone score 69 ± 2; adding the formulas moves that to 68 ± 2. The formulas with no signal samples score 51 ± 2, a coin flip. On a first visit the tumour does not change the scattering spectrum, the frequency slope or the compactness more than healthy fibroglandular tissue does. Comparing the bright antennas with the quiet ones inside the same scan, and the group-delay roughness of the brightest antenna, also sat near 52/100, and that check saw every fair scan at once. The weak first-visit result still lives in the individual time samples, which a 15-number summary throws away. Newest phantoms stay ahead of older ones (73 ± 2 against 56 ± 2).

Follow-up keeps the published logistic model on the rotationally cleaned difference signal. Physics features are computed on that same difference, so a gain on the first visit is not allowed to spoil the follow-up.

Follow-up (246 fair scans)	Difference signal	Difference + physics
Tumour score /100	97 ± 0	98 ± 0
Older / newest phantoms	88 ± 0 / 98 ± 0	88 ± 1 / 99 ± 0
2 cm or smaller	97 ± 0	98 ± 0

On the physics-only logistic model, a positive weight raises the tumour score. The six largest standardised weights, averaged over the training folds, are: spec_entropy +0.35; lambda2 -0.32; slope_r2 -0.24; delay_std_ns -0.21; born_log +0.18; effective_rank +0.14.

Closed-form range and bearing, on the same fair tumour scans: average miss 2.6 cm (44% within 2 cm). gen1 3.8 cm, gen3 2.2 cm. The trained-free image spot on those scans misses by 1.8 cm. The formula is the identification baseline; the image remains the localiser where it already works.

10. Type and criticality

The dataset records shape (sphere or spiculated) and diameter. It does not record benign versus malignant, and its BI-RADS number is breast density. Type here means that shape. Criticality is a research index from the Born volume law:

C = (d / 2 cm)^3, multiplied by 1.5 when the shape is spiculated.

A 2 cm sphere has C = 1 (low). A 2 cm spiculated inclusion has C = 1.5 (moderate). A 4 cm sphere has C = 8 (still moderate). Anything above 8 is high. Healthy scans are not scored. Only 6 of 134 fair tumour scans are spiculated, all in the older phantoms, so a separate shape classifier is not reported: the sample cannot support one. Criticality is reported because it is defined for every tumour scan.

Balanced accuracy (/100)	Signal only	Signal + physics	Chance
Size class, first visit	44 ± 3	45 ± 3	33
Criticality, first visit	71 ± 3	71 ± 3	33
Criticality, follow-up	57 ± 0	56 ± 0	33

Balanced accuracy gives each class the same weight, so a model cannot look good by always saying "small" or "low". Chance for three classes is 33. Follow-up criticality is uneven: older phantoms 36 ± 2, newest phantoms 92 ± 0 correct. The balanced accuracy is the number for the whole follow-up set.

11. Comparison with published studies

These papers answer related questions on different data and under different rules. The point of the table is where this study adds a result they did not report: a first visit and a follow-up kept separate, a phantom that is never in both training and test, and a criticality index on measured scans rather than on a simulated tumour response.

Study	What they reported	What was different
Reimer, Krenkevich and Pistorius, EuCAP 2020	Detection: Accuracy 85 ± 4%	Subset accuracy. The later leave-one-shell-out study found logistic regression did not generalise.
Reimer and Pistorius, IEEE J-ERM 2022	Detection: ROC AUC 78 ± 3%; 90 ± 3% if the test breast volume is inside the training range and the tumour is at antenna height	First-visit detection. Logistic regression and a dense network did not generalise under the same split. No follow-up subtraction and no size or criticality task.
Reimer and Pistorius, Sensors 2023 (review)	Detection from images: Sensitivity 63–100% across 11 studies; specificity 20–65% in the 4 studies that reported it	Specificity is rarely reported. The review treats most machine-learning estimates other than the 2022 leave-one-shell-out result as likely optimistic.

Study	What they reported	What was different
Davis et al., IEEE TBME 2008	Size and shape: Size accuracy above 97%; shape accuracy above 70%	Simulation with a known tumour response and added noise. No skin, fat shell or measured instrument drift.
Conceição, O'Halloran, Jones and Glavin, PIER 2010	Size, shape, then benign versus malignant from shape: Coarse size 93.05% (LDA); coarse shape 80.90% (LDA); best size-then-shape cascade 81.10% (LDA)	Simulation. Benign versus malignant is assigned from the sphere's surface, not from a measured pathology label.
This study	First visit, signal + physics, ROC score 68 ± 2/100. Follow-up 98 ± 0/100. Criticality balanced accuracy 71 ± 3.	Experimental UM-BMID. Fair sessions only. Same phantom never in training and test. Not a clinical benign/malignant label.

Reimer and Pistorius (IEEE J-ERM, 2022) remain the strict published detection number on this dataset: a convolutional network reached 78 ± 3 under leave-one-fat-shell-out, and 90 ± 3 only when the test breast was inside the training volume and the tumour sat at antenna height. Their logistic regression did not generalise under that split. The EuCAP 2020 accuracy of 85 ± 4 was a subset and is not the same test. Davis et al. (2008) and Conceição et al. (2010) show that size and shape can be read from a radar signature when the tumour response is simulated; those accuracies are not available as a target on UM-BMID, which has skin, fat, fibroglandular tissue and only six spiculated tumours in the fair set. Image-based microwave studies reviewed in 2023 report sensitivity from 63 to 100 percent, but specificity, when it is reported at all, is 20 to 65 percent.

This study does not beat the 2022 network on the network's own split, and the new formulas do not beat the signal forest on a first visit. What it adds is the split between a first visit and a follow-up in which both classes are given the same number of earlier healthy scans, the follow-up score near 98/100 under that rule, a checked statement that the Born slope and the scattering spectrum are clutter on a first visit, and a criticality index that can be audited because the diameter and the shape are known. On these measured scans that reading is only partly available: criticality clears chance, and a three-way size label does not.

12. Limits

• Few phantoms (27 single-scan, 24 follow-up), so numbers can move by several points (see ±).

• Follow-up uses 46 older-phantom scans and 200 newest-phantom scans. Generation 2 is excluded on purpose.

• Large (6 cm) tumours exist only in gen1; gen3 has only four tumour positions.

• The gen3 antenna layout was fitted on tumour positions; the gen3 image-spot figure uses the half-and-half check, the all-tumours figure uses the final layout and may be slightly flattering.

• Size and diameter may borrow a little from the recording session, as position did; the "recording details only" column shows that ceiling.

• No benign vs malignant labels exist in this dataset. "BI-RADS" in the metadata is breast density, not cancer risk. Criticality in this report is the Born volume index defined in section 10, not a clinical category.

The sections above are the measured results. The steps below are the internal process, in the order the code runs, with the formula and the number that formula produced.

Module — Tumour classification (the learned reading)

Steps 1-44 turn one file into a picture and a rule-based YES or NO (confidence at least 0.45). That rule does not learn from other phantoms. On the fair UM-BMID test it scores about 50/100, which is a coin flip. This module is the separate learned reading. It asks three questions of phantoms it has never seen: is there a tumour, how big is it, and where is it. Official first-scan score: **71 +/- 2/100**. Official follow-up score: **97 +/- 0/100**. Neither number is a medical diagnosis.

Step 45. Two situations, two subtractions

Input: one UM-BMID scan S(f, antenna), shape (1001, n_antennas), f = linspace(1 GHz, 8 GHz, 1001), plus the empty-chamber scan of that sitting.
First visit has no earlier healthy scan of the same breast. Only the empty chamber is removed.
Follow-up also subtracts earlier healthy scans of the same phantom and the same sitting.

```
S_cal = S - S_empty
S_rot = S_cal - mean_over_antennas(S_cal)
S_follow = S_cal - mean(S_healthy_1 ... S_healthy_k)
```

The rotational mean is the part that looks the same from every antenna: skin, fat shell, chamber. A compact inclusion does not look the same from every angle, so it survives in S_rot. S_follow is stronger when a previous healthy scan exists, because that scan carries the fibroglandular tissue as well as the shell. That is why the follow-up score is 97 +/- 0/100 and the first-scan score is 71 +/- 2/100.

Step 46. Breast time window

The frequency samples are turned into time with an IFFT. The delay of sample n is t_n = n / (N * df). Echoes before 0.71 ns are the antenna. Echoes after 5.71 ns are past the breast. This is the same window used by the published logistic-regression baseline on this dataset.

```
dt = 1 / (N * mean(diff(f)))
keep samples with 0.71e-9 <= t <= 5.71e-9
about 36 time samples on the 1-8 GHz, 1001-point grid
```

Step 47. Signal features (prefix sig_ and rsig_)

Input: S_cal for sig_, and S_rot for rsig_. No tumour coordinate is used.
Output: one number per formula below. The forest and the logistic model see only these numbers, not the raw matrix.

```
cal_to_raw = log10( sum|S_cal|^2 / sum|S|^2 )
rot_residual = sum|S - mean_antenna|^2 / sum|S|^2
For each 1 GHz band from 1-2 through 7-8:
band_log = log10(mean power)
band_cv = std(power per antenna) / mean(power)
band_rot = rotational energy in that band / band energy
For each kept time sample k:
w_log = log10(mean over antennas of |s(t_k)|^2)
w_cv = std / mean of that time sample
Angular harmonics h1..h4 = |FFT of antenna power|[k] / |FFT|[0]
ant_cv = std(antenna energy) / mean(antenna energy)
ant_max_over_median = max(antenna energy) / median(antenna energy)
```

sig_ is the empty-chamber-cleaned scan. rsig_ is the same list after the rotational mean is removed. The breast-only candidate used later keeps just the w_ numbers (the time samples inside 0.71-5.71 ns) and drops the frequency-band numbers, because those bands still contain the antenna reflection.

Step 48. Image features (prefix img_ and imgl_)

Coherent delay-and-sum is run on S_rot. The grid is 64 by 64 over +/- 6 cm. Wave speed and antenna layout are the fitted values for that generation (gen1: 2.7e8 m/s, gen3: 2.1e8 m/s). imgl_ uses only 1-4 GHz, which was the band that localised gen3 best. DMAS-D4 features are stored but are not allowed into the model search: they were built before a baseband fix.

```
peak_over_mean = max(image) / mean(image)
ccr, FWHM_cm, entropy, top-1-percent / median
peak (x, y) of a lightly smoothed image, and its radius
From the brightest blind ROI (no ground truth):
equivalent diameter = 2 * sqrt(area / pi)
aspect, compactness = area / bounding box
eccentricity = (aspect - 1) / (aspect + 1)
local contrast, ring excess, suspicion, rule confidence
The rule confidence is a feature. It is not the classification answer.
```

Step 49. Physics summaries (the side check)

These 15 numbers describe the rotational residual with formulas, instead of handing every time sample to the forest. They were tested after the official model. They did not replace it.

```
Breast-window matrix B (time by antenna), singular values s_i
L_i = s_i^2 / sum(s_j^2)
effective rank = 1 / sum(L_i^2)
On 2-6 GHz, p(f) = mean over antennas of |S_rot|
log10 p(f) = a + b * log10(f / 1 GHz) (ordinary least squares)
A small sphere in the Born approximation has b near 2.
Excess kurtosis of the energy-weighted delay
Angular resultant R = |sum p_k exp(i theta_k)| / sum p_k
Radar range of the brightest antenna:
d = R_antenna - c * t_peak / 2
```

Measured on the same fair first-visit scans. Signal features alone in this narrower test: 69/100. Signal plus these formulas: 68/100. Formulas alone: 51/100. The official nested first-scan score remains 71 +/- 2/100, because that test is allowed the image features and the extra training scans. On a first visit the tumour does not move these summaries more than healthy fibroglandular tissue does.

Step 50. The fair test set

Why. Tumour scans and healthy scans were often taken in different sittings, and the hardware drifts. A model can look clever by recognising the sitting. **Rule.** Keep only sittings that contain both classes. Inside each sitting, draw the same number of tumour scans and healthy scans (the smaller of the two counts, random seed 0). **Resulting test:** 268 first-scan scans from 27 phantoms (134 tumour). Follow-up: 246 scans from 24 phantoms. Generation 2 is left out of the follow-up test: its tumour scans and healthy scans were different sittings, so the sitting number alone gives the answer.

A second model sees only antenna radius, sitting number and generation. On the first-scan test it scores 47/100. On the follow-up test it scores 49/100. Both must stay near 50. If they do not, the sitting is leaking the label.

Step 51. Phantom-grouped nested testing

Input: the fair scans, grouped by phantom name.
Technique: StratifiedGroupKFold. A phantom is never in both the training fold and the test fold. Outer loop: 5 folds. Inner loop, on the training phantoms only: 4 folds. Repeated 3 times with different seeds.
What is chosen inside the inner loop, and nowhere else: which feature family, which model, which training pool, and the alarm cut-off.

```
Feature families:
signal = sig_
signal_rot = rsig_
breast = time-window samples only (sig_w and rsig_w)
image = img_ and imgl_
```

```
all = signal + signal_rot + image
Models: logistic, or a forest of 400 trees
Pools: fair scans only, or fair scans plus every other scan of phantoms that are not in the test fold
The test fold is scored once, with the choice the inner loop made. The reported +/- is the spread of that score across the 3 repeats.
```

What the inner loop picked most often. First scan: all|random_forest|fair_plus_other. Follow-up: signal_rot|logistic|fair. The saved models in results/classification/ are those choices, fit afterwards on all training phantoms. The scores above are the nested test, not the fit on every phantom.

Step 52. The two learners, written out

```
Logistic, used for the official follow-up:
z_j = (x_j - mean_j) / std_j (means and stds from the training phantoms)
p = 1 / (1 + exp(-(b0 + sum b_j z_j)))
penalty: inverse strength C = 0.5, class_weight = balanced
balanced means each class is given the same total weight, so a larger class cannot dominate

Random forest, used for the official first scan:
400 trees, a leaf must hold at least 3 scans
class_weight = balanced, random_state = 0
the score is the fraction of trees that vote tumour

Diameter uses ridge regression (penalty alpha = 3) or a forest regressor (400 trees, min leaf 3, max_features 0.33). Size class uses the same logistic or forest, with three labels.
```

Step 53. The yes/no cut-off

The model outputs a probability p. A scan is called tumour when p is at least t. t = 0.5 is reported, but it is not the official alarm. The official t is chosen on the training phantoms only, at the point where the true-positive rate minus the false-positive rate is largest (the Youden point).

```
t = argmax over thresholds of (TPR - FPR)
TPR = tumours found / all tumours
FPR = healthy scans called tumour / all healthy scans
sensitivity = TPR, specificity = 1 - FPR
```

Saved first-scan cut-off: 0.507. At that cut the nested test finds 66% of tumours and clears 65% of healthy scans. At the default 0.5 it finds 74% and clears 57% (more tumours, more false alarms). Saved follow-up cut-off: 0.663. It finds 88% and clears 95%.

Step 54. How a follow-up reference is chosen

Inside one phantom and one sitting, list the healthy scans. For every scan in that sitting, tumour or healthy, the other healthy scans are the allowed references. Let m be the smallest number of other healthy scans any one of those scans has. Use k = min(3, m). If m is 0, the sitting is dropped. The k scans are drawn with a random generator seeded by the scan id, the same way for both classes. The reference is their average. A tumour is never given a cleaner baseline than the healthy scans, so k cannot give the label away.

Measured. Follow-up score 97 +/- 0/100 (older phantoms 86, newest 99, tumours of 2 cm or smaller 97). The change in the scan grows with tumour diameter on the newest phantoms, which is the check that the subtracted signal is the tumour and not a processing accident. A 1 cm tumour changes the scan about as much as repeating a healthy scan.

Step 55. Size, diameter, type and criticality

```
size = small if d <= 2 cm
medium if d <= 4 cm
large otherwise
Balanced accuracy = average of the three class recalls.
```

```
Chance for three equal classes = 1/3.
Diameter error = mean |predicted cm - true cm|

Type would be spherical versus spiculated. Only 6 of 134 fair tumour scans are spiculated, all in
the older phantoms, so no shape classifier is reported.

Criticality is a research index, not clinical BI-RADS (the metadata BI-RADS number is breast
density):
C = (d / 2 cm)^3
C = 1.5 * C if the shape string contains spiculated
low if C <= 1, moderate if C <= 8, high if C > 8
A 2 cm sphere is low (C = 1). A 2 cm spiculated inclusion is moderate (C = 1.5). A 4 cm sphere is
moderate (C = 8). A 6 cm sphere is high (C = 27).
```

First-scan size class: 64 +/- 3/100 balanced accuracy (recording-details-only 57). First-scan diameter error: 0.7 cm (older 1.1 cm, newest 0.6 cm). Follow-up size class: 56 +/- 1/100, and 90% correct on the newest phantoms. Follow-up diameter error: 0.4 cm, newest phantoms 0.2 cm. The volume-index classifier on the first-scan fair tumours scores 71 +/- 3/100 balanced accuracy (chance 33). Adding the physics summaries leaves it at 71/100.

Step 56. Where the tumour is

A learned position model is trained during the nested test and then not saved. Inside one sitting the tumour sits in the same place, so a model that can tell sittings apart places the tumour without reading the image. The honest localiser never sees a label or a sitting: it is the brightest spot of the 1-4 GHz coherent image (imgl_peak_x_cm, imgl_peak_y_cm).

```
miss_cm = hypot(x_peak - x_true, y_peak - y_true)
radar alternative, not used as the official localiser:
d = R_antenna - c * t_peak / 2
(x, y) = d * (cos theta, sin theta) at the brightest residual antenna
```

On the fair tumour scans the image spot misses by 1.76 cm on average (65% within 2 cm). The nested trained position model misses by 2.04 cm, which is worse, and it is the one that can borrow the sitting. The closed-form radar range, on the same 133 tumour scans, misses by 2.57 cm (44% within 2 cm). The image spot on those scans misses by 1.76 cm (65% within 2 cm). On gen3 the radar range misses by 2.19 cm (48% within 2 cm). Gen1 images still lose to guessing the centre: the dense inner tissue is off-centre and brighter than the tumour. Gen3 is the generation where the image spot is the result to quote. The gen3 antenna layout was fitted using tumour positions, and the held-out half is the number that should be quoted for that layout.

Step 57. What happens for one new scan

Input: the scan, its empty-chamber scan (or the follow-up healthy average), antenna radius, generation.
Steps inside predict.py: S_cal, S_rot, signal features, coherent DAS, low-band DAS, image features, physics summaries.
Then: the saved forest (first scan) or the saved logistic model (follow-up) turns that vector into p. Compare p with the saved Youden cut-off. Size class and diameter come from their own saved models. Criticality is computed from the predicted diameter with the spherical factor, unless a shape string is supplied.
Output: tumour probability, yes/no, size class, diameter in cm, criticality, and the image-spot (x, y). The image spot is marked reliable only for gen3.

Step 58. How the 71 and the 69 fit together

71/100 is 71 +/- 2: nested test, feature family chosen inside the training phantoms, and that choice was signal plus image, random forest, trained with the extra scans. 69/100 is the later side check: the forest was locked to signal features and to the fair training scans only, before the formulas were added. Adding the formulas made 68. Same phantoms, smaller menu of inputs. The 69 did not replace the 71. Oldest-phantom first scans are the weak part (67/100). Newest-phantom first scans are 72/100.

Follow-up on those same newest phantoms is 99/100.

Step 59. What was tried and did not raise the first scan

Pasting a tumour echo measured on a training breast onto other training breasts. The fair test breasts were never touched. The forest score fell.

Subtracting an average fat shell built from other breasts. That removed the tumour as well as the shell, and the first visit stayed with the empty chamber only.

Removing the strongest shared patterns (SVD) and cutting the early skin echo. The shared-pattern removal took the tumour with it. The time cut changed nothing.

The physics summaries in Step 49. They describe the scan, and they do not separate tumour from healthy on a first visit.

A convolutional network on the breast-window sinogram, which is the input behind the published 78 ROC figure, and a PCA-then-logistic classifier on that same sinogram, which is the simulation papers' step. On this fair phantom-held-out test the network scored 50/100 and the PCA logistic 45/100. The network could fit the breasts it trained on (about 71/100 on those breasts) and the score fell to about 58/100 on a held-out training phantom, then to 50/100 on the fair test. Neither replaced the forest.

Step 60. Scores next to the step that produces them

```
Step 45-48 features, nested, first scan: 71 +/- 2/100 (Step 51-53).
Step 45 follow-up subtraction, nested: 97 +/- 0/100 (Step 54).
Step 38 rule checker, same first-scan scans, no training: 50/100.
Step 53 cut-off, first scan: find 66%, clear 65%.
Step 55 size, first scan / follow-up: 64 +/- 3 / 56 +/- 1 balanced accuracy out of 100.
Step 55 diameter, first scan / follow-up: 0.7 cm / 0.4 cm.
Step 56 image spot: 1.8 cm average miss.
Sinogram network / PCA logistic, same fair first-scan test: 50/100 / 45/100. Not adopted.
Files: results/classification/final_summary.json holds every nested number.
results/reports/classification/guide_progress.pdf is the short sheet for a project guide.
```

Full numbers: results/classification/final_summary.json, math_analysis.json, geometry_calibration.json and background_removal_comparison.json.